

Lab Exercise 8

Objectives:

At the end of the lab exercise, student should be able to:

- **Apply global variables for data sharing across functions**
 - Students will practice using global variables to store and access values across multiple functions, highlighting how data can be managed without parameters.
- **Develop structured problem-solving skills through modular design**
 - By separating tasks into input, processing, and output functions, students will strengthen their ability to design programs systematically and improve code readability.
 -

Task 1 : Function with parameter

Write a program that defines **three separate functions with arguments** to perform arithmetic operations on three integers entered by the user:

1. `addNumbers(int a, int b)` → returns the sum of the first two numbers.
2. `multiplyNumbers(int sum, int c)` → returns the product of the sum and the third number.
3. `subtractNumber(int product, int c)` → returns the result of subtracting the third number from the product.

In the `main()` function:

- Prompt the user to enter three integers.
- Call the three functions in sequence.
- Display the results of addition, multiplication, and subtraction.

Sample output

```
Name    : Baharudin Osman.
ID No   : AIU12345678.
Course  : Bac. of Computer Science.

Enter three integers      : 4  23 45
Sum of first two numbers  : 27
Product with third number : 1215
Subtract third number     : 1170
```

Task 2: Using Function and Global variable

Write a program that defines **three functions without parameters**:

1. `inputNumbers()` → asks the user to enter three integers and stores them in global variables.
2. `processNumbers()` → performs the following operations using the stored numbers:
 - Add the first two numbers.
 - Multiply the sum by the third number.
 - Subtract the third number from the product.
3. `displayResults()` → prints the results of addition, multiplication, and subtraction.

Call these functions in sequence inside `main()`.

Sample Output

```
Enter three integers      : 5 6 8
Sum of first two numbers : 11
Product with third number: 88
Subtract third number    : 80
```

Task 3 : Function and Looping

Modify Task 1 so that it uses a looping statement which repeatedly executes until the user chooses to stop. The program must prompt the user during each iteration to press the character 'N' (uppercase) if they wish to terminate the loop. If the user enters any **other character**, the loop should continue running.